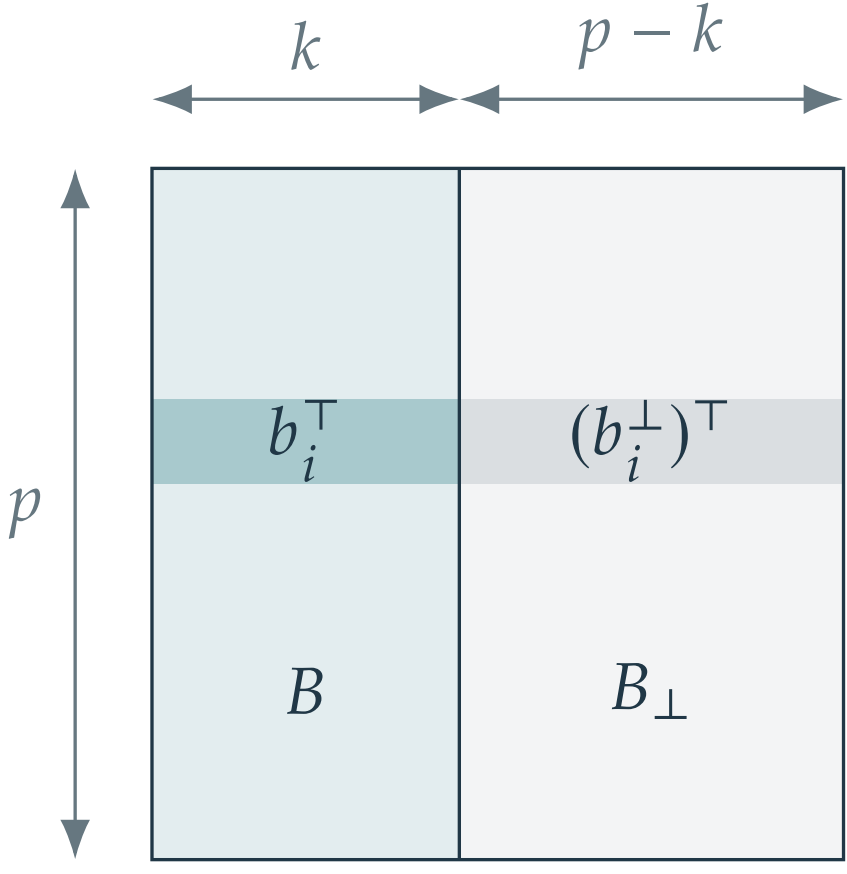

$$\widetilde{B} = [B\ B_\perp], \quad \widetilde{B}\widetilde{B}^\top = I_p$$

$$\|b_i\|^2 + \|b_i^\perp\|^2 = 1 \quad \Rightarrow \quad \beta_i \leq 1$$